AIR POLLUTION AND CONTROL

Subject Code : 23A01504b

UNIT – IV : CONTROL OF GASEOUS POLLUTANTS

Syllabus: Process and equipment for the removal by chemical methods - Design and operation of absorption and adsorption equipment - Combustion and condensation equipment.

I. Introduction – Control of Gaseous Pollutants

Gaseous pollutants (SO_2, NO_x, H_2S, HCl, HF, Cl_2, NH_3, CO, VOCs, hydrocarbons) cannot be removed by mechanical filters. They are controlled by **mass transfer** or **chemical conversion**.

Source control (change fuel / process) → Gas cleaning (end-of-pipe) → Tall stack dispersion (last resort)

Method	Principle	Typical use
Absorption (scrubbing)	Gas dissolves in liquid, physically or with reaction	SO_2, HCl, NH_3, H_2S, NO_x
Adsorption	Gas molecules stick to the surface of a solid	VOC, solvent vapour, odour, Hg
Combustion / oxidation	Burn to CO_2 and H_2O	VOC, CO, odours, HC
Condensation	Cool / compress to liquefy vapour	High-conc. solvent vapour
Chemical conversion (catalytic)	Convert to harmless product using catalyst	NO_x (SCR), CO / HC (converter)

Method	Principle	Typical use
Biofiltration	Micro-organisms degrade pollutants	Odours, low-conc. VOC, H_2S

Selection factors: pollutant type and concentration, gas flow and temperature, required efficiency, recovery value, waste generated, capital and operating cost, safety (flammability), space.

2. Removal by Chemical Methods

In chemical methods the pollutant is converted to a harmless or valuable product by chemical reaction with a reagent (alkali, ammonia, lime, oxygen) in a reactor / scrubber.

2.1 Control of sulphur dioxide - Flue Gas Desulphurisation (FGD)

- **Non-regenerative (throwaway):** reagent is used once and the product is disposed or sold (gypsum)
- **Regenerative:** reagent is recovered and sulphur by-product (S, H_2SO_4, liquid SO_2) is sold (Wellman-Lord, magnesium oxide, citrate process)

FGD process	Reagent / reaction	Efficiency
Wet limestone	$CaCO_3 + SO_2 + 1/2\ O_2 + 2\ H_2O \rightarrow CaSO_4.2H_2O$ (gypsum) + CO_2	90 - 98%
Wet lime	$Ca(OH)_2 + SO_2 \rightarrow CaSO_3.1/2\ H_2O$; oxidised to $CaSO_4$	90 - 98%
Spray dry (semi-dry)	Lime slurry atomised in hot gas; dry $CaSO_3$ collected in baghouse	70 - 90%
Dry sorbent injection	Lime / sodium bicarbonate injected into duct	30 - 60%

FGD process	Reagent / reaction	Efficiency
Double alkali	NaOH scrubbing; regenerate with lime: Na_2SO_3 + $Ca(OH)_2$ -> $CaSO_3$ + 2 NaOH	90 - 95%
Wellman-Lord	Na_2SO_3 + SO_2 + H_2O -> 2 $NaHSO_3$; heat gives conc. SO_2	90 - 97%

- Other SO_2 methods: fluidised bed combustion with limestone, coal washing, switching to low-sulphur fuels, **ammonia scrubbing** (gives ammonium sulphate fertiliser)

2.2 Control of oxides of nitrogen (NO_x)

- **Combustion modification (primary):** low-NO_x burners, staged combustion, flue gas recirculation, lower excess air and lower peak temperature
- **Selective Catalytic Reduction (SCR):** 4 NO + 4 NH_3 + O_2 -> 4 N_2 + 6 H_2O at **300 - 400 deg C** over V_2O_5/TiO_2 catalyst; 80 - 95% removal; ammonia slip < 5 ppm
- **Selective Non-Catalytic Reduction (SNCR):** NH_3 or urea injected at **850 - 1100 deg C**; 30 - 60% removal; no catalyst
- **Wet scrubbing:** NO is poorly soluble; oxidise (ozone, ClO_2, H_2O_2) to NO_2 then absorb in alkali: 2 NO_2 + 2 NaOH -> $NaNO_2$ + $NaNO_3$ + H_2O
- **Three-way catalytic converter** (vehicles): NO_x reduced, CO and HC oxidised (see Unit V)

2.3 Control of H_2S, HCl, HF, Cl_2, NH_3 and CO

- **H_2S:** absorbed in amine (MEA, DEA) or alkaline solutions; **Claus process** converts it to sulphur: $H_2S + 3/2\ O_2 \rightarrow SO_2 + H_2O$; $2\ H_2S + SO_2 \rightarrow 3\ S + 2\ H_2O$ (Al_2O_3 catalyst)
- **HCl, HF, Cl_2:** highly soluble - water or caustic soda scrubbers: $HCl + NaOH \rightarrow NaCl + H_2O$; $Cl_2 + 2\ NaOH \rightarrow NaCl + NaOCl + H_2O$
- **NH_3:** absorbed in dilute acid (H_2SO_4) - gives ammonium sulphate
- **CO:** oxidised over a Hopcalite (MnO_2-CuO) or noble metal catalyst; $2\ CO + O_2 \rightarrow 2\ CO_2$
- **VOCs:** catalytic oxidation, bio-oxidation, UV photocatalysis

3. Absorption

Absorption is the transfer of a gaseous pollutant (solute) from the gas mixture into a liquid (solvent / absorbent). Physical absorption depends on **solubility**; chemical absorption (reaction in the liquid) greatly increases the rate and capacity.

3.1 Fundamentals

- **Henry's law** (dilute solutions): $p = H\ x$ or $y = m\ x$, $m = H / P$ (p = partial pressure of solute, x = mole fraction in liquid, H = Henry's constant). Low $H \rightarrow$ highly soluble gas (NH_3, HCl); high $H \rightarrow$ poorly soluble (CO, NO)
- **Equilibrium line** ($y^* = m\ x$) and **operating line** (mass balance) decide driving force ($y - y^*$)
- **Two-film theory (Whitman):** resistance to transfer lies in a thin gas film and a thin liquid film at the interface. $1/K_G = 1/k_g + H/k_l$. Rate $N_A = K_G\ a\ (p - p^*)$ per volume

- For **very soluble** gases the **gas film** controls (NH_3, HCl); for **sparingly soluble** gases the **liquid film** controls (CO_2, O_2, NO); chemical reaction removes the liquid-film resistance
- **Absorption factor A = L / (m G)**: if A > 1, high removal is easy

3.2 Choice of absorbent (solvent)

- High solubility for solute and **selective**; low volatility (less solvent loss); low viscosity; non-corrosive, non-toxic, non-flammable, cheap, easily regenerated, stable
- Water for NH_3, HCl, HF, SO_2 (moderate); NaOH / Na_2CO_3 / lime for acid gases; amines for H_2S and CO_2; oils for hydrocarbons

3.3 Absorption equipment (gas scrubbers)

Equipment	Features	Application
Packed tower	Gas and liquid contact over packing surface; counter-current	Most common; > 95% removal for soluble gases
Plate (tray) tower	Sieve / bubble-cap / valve trays; gas bubbles through liquid	Large flows, large liquid loads, fouling liquids
Spray tower / chamber	Liquid sprayed into rising gas; low dP	Highly soluble gases; combined with PM removal
Venturi scrubber	High velocity atomises liquid	Simultaneous particulate + gas
Bubble column	Gas bubbled through liquid pool	Reactive gases, slurry

3.4 Packed tower

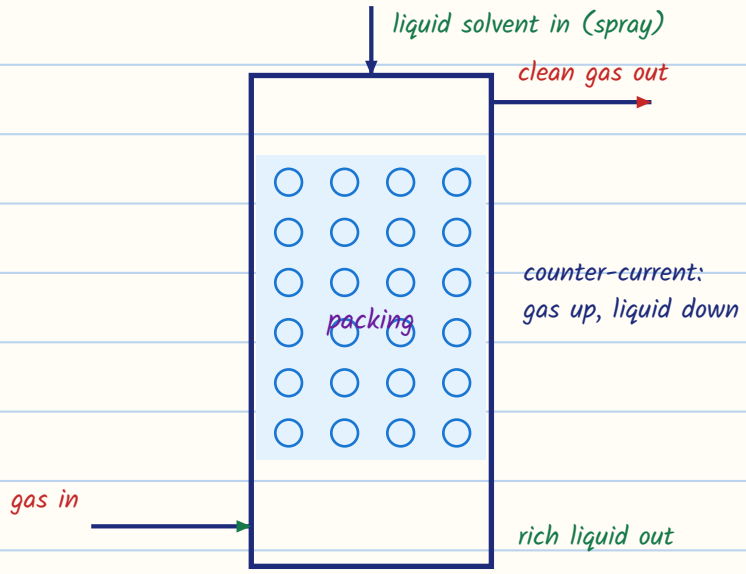


Fig. Counter-current packed absorption tower

- **Shell** (steel / FRP / PVC) filled with **packing** supported on a grid; **liquid distributor** at the top gives uniform wetting; **gas distributor** at the bottom; **mist eliminator** on top; **redistributors** every 3 - 5 tower diameters
- **Packing** gives large surface area, voidage, low pressure drop: **random** - Raschig rings, Berl saddles, Intalox saddles, **Pall rings** (best), Tellerettes; **structured** - corrugated sheets (low dP, costly); materials: ceramic, plastic, metal, carbon
- Packing size: 1/8 th of tower diameter at most; typical size 25 - 50 mm

Design of a packed tower (dilute gas, straight equilibrium line)

1. **Material balance:** $G\,(y_1 - y_2) = L\,(x_1 - x_2)$ (G and L = molar flows of inert gas and pure solvent; use Y and X ratios for non-dilute)
2. **Minimum liquid rate:** $(L/G)_{min} = (y_1 - y_2) / (x_1^* - x_2)$ with $x_1^* = y_1/m$; **actual** $L = 1.2 - 1.5 \times L_{min}$

3. **Tower diameter:** from gas flow and permissible superficial velocity = **50 - 75% of flooding velocity** (from Sherwood-Leva-Eckert generalised pressure drop chart); D = sqrt(4 Q / (pi v)); pressure drop 200 - 400 Pa per metre of packing
4. **Height of packing: Z = H_{OG} x N_{OG}** (height of a transfer unit x number of transfer units)
5. **H_{OG} = G / (K_y a)** (typically 0.3 - 1 m); **N_{OG} = integral dy / (y - y^*)** from y_2 to y_1
6. For a dilute gas with straight lines (Colburn): **N_{OG} = [1 / (1 - 1/A)] ln{ (1 - 1/A) (y_1 - m x_2) / (y_2 - m x_2) + 1/A }**, with A = L / (mG)
7. If x_2 = 0 and A very large, N_{OG} = ln (y_1 / y_2) = ln (1 / (1 - eta))

Example: to remove 99% of a solute with N_{OG} = ln (100) = 4.6 and H_{OG} = 0.6 m -> **Z = 4.6 x 0.6 = 2.8 m** (plus 15-20% safety).

Packed tower: high contact area, low dP, suitable for corrosive gas (plastic) | Plate tower: for large L/G, solids, easy cleaning and cooling coils.

- **Operating problems:** flooding (excess gas / liquid), channelling, foaming, fouling and scaling ($CaSO_4$), loading of dust, **weeping** and dumping of trays
- **Disposal:** spent solution needs treatment (neutralise, settle); some gives by-products (gypsum, ammonium sulphate)

4. Adsorption

Adsorption is the accumulation (adhesion) of molecules of a gas (**adsorbate**) on the surface of a solid (**adsorbent**). It is a surface phenomenon (different from absorption) and is exothermic.

Property	Physical adsorption (physisorption)	Chemical adsorption (chemisorption)
Forces	Van der Waals (weak)	Chemical bonds (strong)
Heat of adsorption	10 - 40 kJ/mol	40 - 400 kJ/mol
Reversibility	Reversible; easy regeneration	Irreversible mostly
Layers	Multilayer	Monolayer
Temperature	Decreases with temperature rise	Increases first, then decreases
Specificity	Not specific	Highly specific

4.1 Adsorbents

Adsorbent	Surface area (m^2/g)	Application
Activated carbon (coal / coconut shell)	600 - 1500	Non-polar VOCs, solvents, odours, Hg, H_2S, dioxins
Silica gel	200 - 800	Dehydration of gases, polar compounds
Activated alumina	200 - 350	Moisture, HF, H_2S, drying
Molecular sieves / zeolites	500 - 800	Selective (pore size 3A, 4A, 5A, 13X), drying, SO_2, CO_2
Polymeric resin, fly ash, lime	Variable	Special organics, acid gases

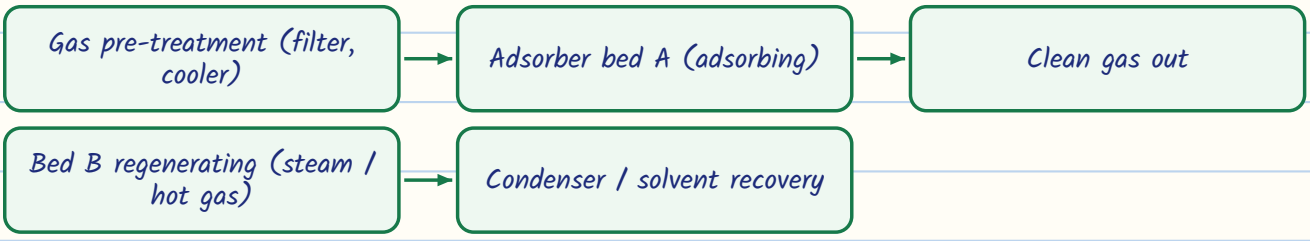
Good adsorbent: large surface area / porosity, high capacity and selectivity, mechanical strength, regenerable, cheap, low pressure drop, no catalytic activity (carbon can ignite with ketones).

4.2 Adsorption isotherms

- Show the amount adsorbed per unit mass of adsorbent (q or x/m) vs equilibrium concentration (or pressure) at constant temperature
- **Freundlich:** $q = K C^{1/n}$ (empirical; log q vs log C is a straight line)
- **Langmuir:** $q = q_m b C / (1 + b C)$ (monolayer; q -> q_m at high C)
- **BET:** multilayer; used to determine the surface area
- Adsorption capacity increases with **higher concentration, higher pressure, lower temperature, larger surface area**; humidity reduces carbon capacity (keep RH < 50%)

4.3 Adsorption equipment

- **Fixed-bed adsorber (most common):** two or more beds - one adsorbs while the other is regenerated (**swing operation**); vertical or horizontal; gas velocity 0.2 - 0.5 m/s; bed depth 0.3 - 1.5 m; 1 - 3 kPa pressure drop
- **Moving-bed / fluidised-bed:** continuous flow of adsorbent, regenerated in a separate vessel; for high flow rate
- **Rotary wheel concentrator:** zeolite wheel adsorbs dilute VOC and desorbs in a small hot stream to an oxidiser
- **Canister / disposable cartridge:** small flows; deodorisation, gas masks



4.4 Breakthrough curve and bed design

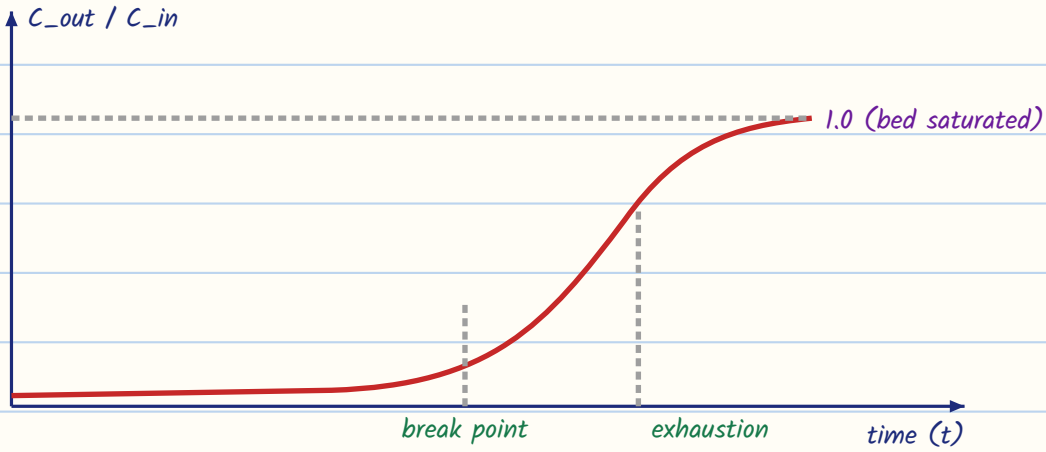


Fig. Breakthrough curve of a fixed bed

- Gas enters the bed; the **mass transfer zone (MTZ)** moves through the bed. When its front reaches the bed end, the outlet conc. begins to rise (**break point**, normally $C/C_0 = 0.05$). At **exhaustion** $C = C_0$ (bed saturated)
- A narrow MTZ means efficient use of the bed. Bed **working capacity** = 30 - 70% of the equilibrium capacity
- **Bohart-Adams / BDST equation:** $t_b = (N_0 / C_0 v) Z - (1 / K C_0) \ln (C_0 / C_b - 1)$ where N_0 = adsorption capacity per volume of bed, v = gas velocity, Z = bed depth, K = rate constant
- **Critical bed depth** $Z_c = (v / K) \ln(C_0 / C_b - 1)$: the minimum Z for any breakthrough time (t = 0)
- **Simple design:** mass of adsorbent = (mass of pollutant to be adsorbed per cycle) / (working capacity); e.g. 100 kg/h solvent x 8 h = 800 kg; with capacity 0.15 kg/kg -> **5300 kg** of carbon (per bed)

4.5 Regeneration

- **Thermal swing:** low-pressure steam (100 - 150 deg C) or hot inert gas drives off adsorbate; steam is condensed and solvent recovered by decanting / distillation; then bed is dried and cooled
- **Pressure swing / vacuum swing (PSA):** reduce pressure to desorb; used for small volume streams
- Chemical regeneration or replacement for chemisorbed gases; carbon lost by attrition, deactivated by high-boiling compounds
- **Advantages:** recovery of valuable solvent, 95 - 99% removal even for low concentrations, handles odours and mixtures, no wastewater (if dry)
- **Disadvantages:** pre-treatment needed (dust, moisture, T < 40 deg C), high cost of adsorbent, fire hazard in carbon beds (hot spots), disposal of spent adsorbent, high energy for regeneration

5. Combustion (Thermal and Catalytic Oxidation)

Organic vapours, CO and odorous gases are **oxidised (burnt)** at high temperature into CO_2 and H_2O: **$C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$ + heat**. Efficiency 95 - 99.99%. Needs good mixing.

The three T's of combustion: TEMPERATURE (650-1100 deg C) + TIME (residence 0.3 - 1 s) + TURBULENCE (good mixing)

Equipment	Operation	Remarks
Direct flame (thermal) incinerator / afterburner	Gas heated to 650 - 1100 deg C with burner; residence 0.5 s	95 - 99%; fuel cost high; halogens give HCl
Recuperative thermal oxidiser	Hot exit gas preheats feed through a heat exchanger (up to 70% recovery)	Lower fuel; fixed cost higher

Equipment	Operation	Remarks
Regenerative thermal oxidiser (RTO)	Ceramic beds store heat; flow switches alternately; 95% heat recovery	Low fuel for dilute VOC; 99% destruction
Catalytic oxidiser	Pt / Pd or metal oxide catalyst; 250 - 500 deg C; lower temperature and cost	Poisoned by S, Pb, P, Hg, Si; 90 - 98%
Flare	Open burning at an elevated or ground-level tip, steam-assisted to avoid smoke	Emergency, waste gas of refineries; 98% destruction; noise, light
Boilers / process heaters	Waste gas used as fuel supplement	Energy recovery with high residence time

Design aspects

- **Residence time:** $V = Q \times t$ (Q at operating temperature; t = 0.3 - 1 s); length : dia ratio about 2
- **Heat balance:** $Q_{fuel} = m\ C_p\ (T_{out} - T_{in})$ (1 - heat recovery) - waste gas heating value can cover the load if VOC conc. is high
- **Safety:** keep VOC concentration below **25% of the lower explosive limit (LEL)** (use dilution air); flame arrestors, explosion vents
- **Heating value rule:** a stream with > 1.9 MJ/m^3 (50 Btu/scf) can sustain combustion without added fuel
- Problems: NO_x formation (thermal NO_x at > 1200 deg C), CO from incomplete burning, acid gases from Cl, S compounds -> scrubber needed after the incinerator

Catalytic converter reaction: $CO + 1/2\ O_2 \rightarrow CO_2$; $C_xH_y + O_2 \rightarrow CO_2 + H_2O$. Catalyst forms: honeycomb ceramic monolith or pellets; space velocity 10,000 - 50,000 per hour; life 3 - 5 years.

6. Condensation

Condensation removes a vapour from a gas stream by **lowering the temperature** (or raising the pressure) until the **partial pressure of the vapour exceeds its saturation vapour pressure**; vapour changes to liquid. It is best for **high concentration (> 5000 ppm) and high boiling** vapours (solvent recovery); often used as pre-treatment before adsorption, absorption or incineration.

Types of condensers

Type	Description	Remarks
Surface condenser (shell and tube / plate)	Coolant (water, brine) flows separately across tubes; condensate collected and does not mix	Product recovered pure; large area; common
Contact (direct) condenser	Coolant (water spray) mixes directly with vapour (spray tower, jet, barometric)	Cheap, compact; condensate contaminated with water
Refrigerated / cryogenic condenser	Chilled coolant (-10 to -40 deg C) or liquid N_2 (-196 deg C)	> 90 - 99% recovery of volatile VOC; costly

Design equations

- **Condensation ends when partial pressure of vapour = vapour pressure p^* at condenser outlet temperature:** mole ratio $Y_{out} = p^* / (P - p^*)$
- **Recovery (efficiency): eta = $1 - Y_{out} / Y_{in}$**; $Y_{in} = y_{in} / (1 - y_{in})$
- **Heat duty:** $Q = m_g C_{pg} (T_1 - T_2) + m_v [C_{pv} (T_1 - T_{dew}) + lambda + C_{pl} (T_{dew} - T_2)]$
- **Heat transfer area:** $A = Q / (U \times DT_{lm})$ with $U = 100 - 500$ $W/m^2.K$ for surface condensers; coolant flow from heat balance
- Antoine equation for vapour pressure: $\log p^* = A - B / (T + C)$

Example: a stream has 10% (v/v) solvent vapour at 1 atm (760 mm Hg) and is cooled to a temperature at which $p^* = 20$ mm Hg. $Y_{in} = 0.10/0.90 = 0.111$; $Y_{out} = 20 / (760 - 20) = 0.0270$; **recovery = $1 - 0.0270/0.111 =$ 75.7%**. To get 90% recovery the p^* must be only 8.5 mm Hg - a lower temperature.

- **Advantages:** solvent can be recovered and reused, no secondary waste in surface type, simple. **Limitations:** inefficient for dilute streams, large cooling load; non-condensable gas still carries vapour (needs afterburner / carbon bed polishing); mist formation

7. Comparison and Selection of Equipment

Method	Gas conc. range	Efficiency	Recovery
Absorption	250 - 10,000 ppm	90 - 99%	Yes (by regeneration)
Adsorption	50 - 10,000 ppm	90 - 99%	Yes (solvent)
Thermal oxidation	100 - 2000 ppm (< 25% LEL)	95 - 99.99%	No (heat only)
Catalytic oxidation	100 - 2000 ppm	90 - 98%	No
Condensation	> 5000 ppm	50 - 90% (99% cryogenic)	Yes (liquid product)
Biofilter	< 1000 ppm	80 - 95%	No

8. Quick Revision / Important Questions

1. Explain the methods of controlling SO_2 emission (FGD). Differentiate regenerative and non-regenerative processes.
2. Explain SCR and SNCR for NO_x control.
3. Describe the principle and design of a packed absorption tower. What is HTU / NTU?
4. Compare physical and chemical adsorption. Explain adsorption isotherms.
5. With a neat sketch explain a fixed-bed adsorber; breakthrough curve and regeneration.
6. Explain thermal and catalytic incinerators and flares.
7. Explain the working of surface and contact condensers; give the design equations.